

# Macroeconomics Notes

Abror Maksudov

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## 1 Gross Domestic Product (GDP)

Gross Domestic Product (GDP) is the total market value of all **final goods and services** produced **within a country's borders** in a **specific time period**.

GPD does not include:

- Intermediate goods
- Non-market activities
- Used goods transactions
- Financial transactions
- Illegal activities
- Environmental externalities

## 2 Methods of Measuring GDP

### 1. Production (Value-Added) Approach:

- Aggregates the value added at each stage of production.
- Value added = Revenue – Cost of intermediate inputs.

### 2. Expenditure Approach:

- Adds up total spending on final goods and services.
- $GDP = \text{Consumption} + \text{Investment} + \text{Government Spending} + (\text{Exports} - \text{Imports})$ .

### 3. Income Approach:

- Sums all incomes earned by households and businesses during production.
- $GDP = \text{Labor income} + \text{Capital income}$ .
- $GDP = \text{Wages} + \text{Profit} + (\text{Rent} + \text{Interest}) + (\text{Indirect Taxes} - \text{Subsidies}) + \text{Depreciation} + \dots$

## 3 Gross National Product (GNP)

Gross National Product (GNP) is the total market value of all **final goods and services** produced **by a country's residents and businesses**, regardless of where that production takes place **during a specific time period**.

- GDP focuses on *where* production happens.
- GNP focuses on *who* produces it (the country's citizens and businesses).

The relationship between GNP and GDP:

$$GNP = GDP + \text{Net Factor Income from Abroad}$$

Where "Net Factor Income from Abroad" represents income earned by domestic residents from foreign investments minus income earned by foreign residents from domestic investments. This means if a country earns more from abroad than foreigners earn domestically, GNP will be higher than GDP, and vice versa.

## 4 National Income Accounting Identity

The relationship between a country's total income and its expenditures:

$$Y = C + I + G + (X - M)$$

Where Y is GDP, C is household consumption, I is business investment, G is government purchases, and (X – M) is net exports.

### 1. Consumption (C)

- *Definition:* All household spending on goods and services, excluding new housing.
- *Includes:*
  - **Durable Goods:** Long-lasting items (cars, household appliances, furniture).
  - **Non-Durable Goods:** Quick-use items (food, clothing, gasoline).

- **Services:** Non-physical, intangible products (healthcare, legal services, education).

## 2. Investment (I)

- *Definition:* Private sector spending on capital goods and changes in inventories.
- *Includes:*
  - **Business Fixed Investment:** Capital expenditures by firms (equipment, factories).
  - **Residential Fixed Investment:** Spending on new housing units by households and landlords (new home constructions and purchases, major home renovations).
  - **Inventory Investment:** Changes in the stock of unsold goods held by firms.
- *Excludes:* Financial assets like stocks, bonds, or cryptocurrencies, as these do not represent production.

## 3. Government Spending (G)

- *Definition:* Government purchases of goods and services.
- *Includes:*
  - Federal, state, and local spending on goods and services.
- *Excludes:* Transfer payments (social security, unemployment benefits, welfare), because these are not payments for goods or services.

## 4. Net Exports (NX = Exports – Imports)

- *Definition:* The difference between exports (X) and imports (M).
- *Includes:*
  - **Exports (X):** Goods and services produced domestically and sold to foreign buyers.
  - **Imports (M):** Goods and services produced abroad and purchased by domestic consumers.
- Can result in trade surplus (positive) or deficit (negative).

# 5 Typical GDP Component Distribution

### Expenditure Approach:

- Consumption (C):  $\sim 70\%$  – *Largest component*
- Investment (I):  $\sim 15\text{-}20\%$
- Government (G):  $\sim 15\text{-}20\%$
- Net Exports (NX):  $\sim \pm 5\%$  – *Smallest component*

### Income Approach

- Wages:  $\sim 50\text{-}60\%$  – *Largest component*
- Profits:  $\sim 20\text{-}30\%$
- Rent + Interest:  $\sim 5\text{-}10\%$
- Indirect Taxes - Subsidies:  $\sim 5\text{-}10\%$
- Depreciation:  $\sim 10\text{-}15\%$

## 6 Disposable Income

Disposable Personal Income (DPI) is the amount of money that households have available to spend or save after paying income taxes: Disposable Income = Gross Income - Taxes.

$$Y^d = Y - T \quad \text{or} \quad Y^d = C + S$$

Where  $Y^d$  is disposable income,  $Y$  is total income,  $T$  is taxes,  $C$  is consumption,  $S$  is savings.

## 7 Net Domestic Product (NDP)

Net Domestic Product (NDP) measures the total value of all final goods and services produced within a country in a given period, after subtracting depreciation.

### 7.1 NDP at Market Prices (NDP-MP)

NDP at Market Prices is valued at the prices paid by buyers, including indirect taxes net of subsidies.

$$\text{NDP-MP} = \text{GDP} - \text{Depreciation}$$

### 7.2 NDP at Factor Cost (NDP-FC)

NDP at Factor Cost is valued at the cost to factors of production, excluding indirect taxes and including subsidies.

$$\text{NDP-FC} = \text{GDP} - \text{Depreciation} - \text{Indirect Taxes} + \text{Subsidies}$$

## 8 Net Domestic Income (NDI)

Net Domestic Income (NDI) measures the total income earned by all factors of production within a country in a given period, after subtracting depreciation.

### 8.1 NDI at Market Prices (NDI-MP)

NDI at Market Prices aligns with market prices, including indirect taxes net of subsidies, and equals NDP-MP.

$$\text{NDI-MP} = \text{GDP} - \text{Depreciation}$$

### 8.2 NDI at Factor Cost (NDI-FC)

NDI at Factor Cost reflects the income to factors, excluding indirect taxes and including subsidies, and equals NDP-FC.

$$\text{NDI-FC} = \text{GDP} - \text{Depreciation} - \text{Indirect Taxes} + \text{Subsidies}$$

## 9 Real vs. Nominal Variable

- **Nominal Variable:** Measured in current market prices, not adjusted for inflation.
- **Real Variable:** Adjusted for inflation, reflecting the true purchasing power.

$$\text{Real Variable} = \frac{\text{Nominal Variable}}{\text{Price Level}} \times 100$$

Where Price Level is an index like CPI, PPI or GDP deflator.

**Purchasing power** is the quantity of goods or services that can be bought with a specific amount of money at a specific time.

$$\text{Purchasing Power (for a specific good)} = \frac{\text{Nominal Income}}{\text{Price of the Good}}$$

## 10 Price Index

A price index is a statistical measure that tracks the average change in prices of a specific basket of goods and services over time, relative to a base period.

**Price Index for a Single Good:**

$$\text{Price Index} = \frac{\text{Price in Current Year}}{\text{Price in Base Year}} \times 100$$

More formally,

$$PI_t = \frac{P_t}{P_0} \times 100$$

Where  $P_t$  is price of good in current year,  $P_0$  is price of good in base year.

**Price Index for a Basket of Goods:**

$$\text{Price Index} = \frac{\text{Cost of Basket in Current Year}}{\text{Cost of Basket in Base Year}} \times 100$$

or,

$$\text{Price Index} = \frac{\sum(\text{Price in Current Year} \times \text{Quantity in Base Year})}{\sum(\text{Price in Base Year} \times \text{Quantity in Base Year})} \times 100$$

More formally,

$$PI_t = \frac{\sum_{i=1}^n (P_{i,t} \times W_i)}{\sum_{i=1}^n (P_{i,0} \times W_i)} \times 100$$

or,

$$PI_t = \sum_{i=1}^n w_i \times \frac{P_{i,t}}{P_{i,0}} \times 100$$

Where:

- $n$  = Total number of goods in the basket.
- $P_{i,t}$  = Price of good  $i$  in current year.
- $P_{i,0}$  = Price of good  $i$  in base year.
- $W_i$  = Weight (proportion of spending) for good  $i$ .
- $w_i = \frac{W_i}{\sum W_i}$ .

**Laspeyres Price Index:**

$$PI_t^L = \frac{\sum_{i=1}^n (P_{i,t} \times Q_{i,0})}{\sum_{i=1}^n (P_{i,0} \times Q_{i,0})} \times 100$$

Where  $Q_{i,0}$  is quantity of good  $i$  in base year.

## 11 Changes in Nominal and Real GDP

1. **Nominal GDP** =  $\sum (P_t \times Q_t)$ :

- *Definition:* Value of output in current prices.
- Changes in nominal GDP can be due to:
  - **Changes in Prices** ( $P_t$ ): Inflation or deflation.
  - **Changes in Quantities** ( $Q_t$ ): More or less output produced (real growth or decline).

2. **Real GDP** =  $\sum (P_0 \times Q_t)$ :

- *Definition:* Value of output in constant base-year prices.
- Changes in real GDP can be due to:
  - **Changes in Quantities Only** ( $Q_t$ ): Prices are fixed, so only output changes matter.

## 12 Calculating Nominal and Real GDP

Recall that  $\text{GDP} = P_1Q_1 + P_2Q_2 + \cdots + P_nQ_n$

**Nominal GDP:** Nominal GDP is the total monetary value of final goods and services produced in year  $t$ , using current prices:

$$\text{Nominal GDP}_t = \sum_{i=1}^n P_{i,t} \times Q_{i,t}$$

**Real GDP:** Real GDP measures the value of production in year  $t$  using constant prices from a base year  $b$ :

$$\text{Real GDP}_t = \sum_{i=1}^n P_{i,b} \times Q_{i,t}$$